

FINITE GALOIS DESCENT IN TELESCOPIC ALGEBRAIC K -THEORY

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ABSTRACT. Fix a prime p and an integer $n \geq 0$. We prove that every finite G -Galois extension $R \rightarrow S$ of $T(n)$ -local commutative ring spectra satisfies

$$L_{T(n+1)}K(R) \simeq (L_{T(n+1)}K(S))^{hG}.$$

For $n \geq 1$, we establish prime-to- p cyclic descent by a comparison of crossed-product Hochschild sectors, realized by a morphism of cyclotomic spectra. For $n = 0$, we prove that even invertible modules of order prime to p have class 1 in the zeroth homotopy group of height-one localized K -theory of arbitrary rational rings. This follows from invariance under affine torsors for even invertible modules and a finite cell argument. The family-completeness theorem of Clausen–Mathew–Naumann–Noel then gives descent for every finite group in both cases.

1. INTRODUCTION

Fix a prime p . For $n \geq 1$, let $T(n)$ denote the telescope of a v_n -self map of a finite p -local spectrum of type n , and let $T(0) = H\mathbb{Q}$. We use nonconnective algebraic K -theory and write

$$K(A) = K(\mathrm{Perf}(A)).$$

Here $\mathrm{Perf}(A)$ is the category of ordinary perfect A -modules. Tensor products in the category of $T(n)$ -local spectra are denoted by

$$X \widehat{\otimes} Y = L_{T(n)}(X \otimes Y).$$

Relative tensor products of local modules use the same convention. All actions and limits are homotopy coherent.

Definition 1.1. A $T(n)$ -local G -Galois extension, for a finite group G , is a map $R \rightarrow S$ in $\mathrm{CAlg}(\mathrm{Sp}_{T(n)})$ with an action of G on S by commutative R -algebra automorphisms such that the canonical maps

$$R \xrightarrow{\simeq} S^{hG}, \quad S \widehat{\otimes}_R S \xrightarrow{\simeq} \prod_{g \in G} S, \quad x \otimes y \mapsto (xg(y))_{g \in G} \quad (1.1)$$

are equivalences.

Theorem 1.2. Let $n \geq 0$, and let $R \rightarrow S$ be a $T(n)$ -local G -Galois extension of commutative ring spectra, with G finite. Extension of scalars induces an equivalence

$$L_{T(n+1)}K(R) \xrightarrow{\simeq} (L_{T(n+1)}K(S))^{hG}. \quad (1.2)$$

For $n = 0$, the inputs are arbitrary rational commutative ring spectra. At positive height the Galois tensor product is local, whereas the K -theory in the conclusion is defined by ordinary perfect modules. Proposition 2.6 relates these two module categories. In particular, ordinary perfection of S over R is not an additional hypothesis of the theorem.

The positive-height finite Galois descent conclusion was announced in work of Burklund and Clausen, as discussed in [Cla24]. The argument here uses the chromatic descent and vanishing results of Clausen–Mathew–Naumann–Noel [CMNN20, CMNN] and the purity theorem of Land–Mathew–Meier–Tamme [LMMT24]. The announcement itself is not used as a proof input.

The proof. The proof of [CMNN, Theorem 6.5] gives completeness of equivariant localized K -theory for the family of cyclic subgroups of order prime to p . Once the canonical comparison is known for these subgroups, family detection proves (1.2); this argument is given in Section 7. Thus the remaining assertion is cyclic descent.

For $n \geq 1$, induction on the order reduces cyclic descent to C_ℓ , with $\ell \neq p$ prime. Form the colimit C of the C_ℓ -action on $\tau_{\geq 0}S$ in commutative algebras. The Galois identities imply $L_{T(n)}C = 0$. For each nonidentity $g \in C_\ell$, the colimit unit identifies the g -twisted Hochschild homology of $\tau_{\geq 0}S$ with $\mathrm{THH}(C)$. Sections 3 and 4 show that these maps detect an equivalence between the cofibers of the cyclotomic assembly maps for $\tau_{\geq 0}S$ and C . Chromatic vanishing kills the latter cofiber after the cyclotomic trace. This proves the K -theory assembly equivalence; induction–restriction and the comparison of module categories then identify it with Galois descent.

For $n = 0$, the needed assertion is instead

$$[L] = 1 \quad \text{in } \pi_0 L_{T(1)}K(R)$$

for every invertible module L with trivial self-braiding and finite order prime to p over a rational ring R . Theorem 5.2 proves this assertion. Its proof constructs a finite sequence of commutative algebra extensions ending at zero. Killing an odd section of the transported line preserves $\mathbb{A}^1$ -invariant localizing invariants; killing an even section gives a perfect extension whose transfer composite is $1 - [L]^{-1}$. On the complementary averaging summand for L , that Euler class is invertible. Each map in the finite sequence is therefore split injective on this summand, which must vanish. After adjoining roots of unity, character decomposition and transfer give tame cyclic descent. Descent along the finite free coefficient extension removes the roots of unity. Section 6 completes this argument.

Conventions. Write $\mathrm{Sp}_{(p)}$ for p -local spectra. Unless indicated otherwise, tensor products and colimits are derived and are formed in spectra. Limits of local spectra are computed in spectra. Localization is symmetric monoidal as a functor into the local category and preserves colimits of commutative algebras. Functoriality with respect to group actions includes the residual quotient actions. A finite group is called *tame* when its order is prime to the fixed prime p . We allow the zero unital algebra.

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2. GALOIS EXTENSIONS AND LOCALIZED PERFECT MODULES

Lemma 2.1 (Averaging). *Let H have order m , prime to p . For a p -local spectrum V with H -action, both V_{hH} and V^{hH} are naturally retracts of V , with image the averaging idempotent*

$$e = \frac{1}{m} \sum_{h \in H} h.$$

The norm $V_{hH} \rightarrow V^{hH}$ is an equivalence. The same conclusions hold in a presentable stable category on objects on which multiplication by m is invertible. An exact functor between stable categories preserves these homotopy orbits and homotopy fixed points whenever they exist.

Proof. The central idempotent $e \in \pi_0 \mathbb{S}_{(p)}[H]$ splits the regular left and right $\mathbb{S}_{(p)}[H]$ -modules. The augmentation restricts to an equivalence

$$e\mathbb{S}_{(p)}[H] \xrightarrow{\cong} \mathbb{S}_{(p)}$$

of $\mathbb{S}_{(p)}[H]$ -modules with trivial action on the target. To check this, use $\pi_* \mathbb{S}_{(p)}[H] = \bigoplus_H \pi_* \mathbb{S}_{(p)}$: on the image of e , the augmentation is an isomorphism in every degree. Thus the trivial module is a retract of the regular module. Tensoring with it or mapping out of it computes, respectively, homotopy orbits and homotopy fixed points, and gives the asserted retract descriptions.

The norm is the sum of the action maps. On the averaging summand it is multiplication by m , hence invertible. The argument works for objects of a stable category by its spectral action, replacing the sphere by the sphere with m inverted. These retracts are formed from finite sums and a split idempotent. Exact functors preserve finite sums and split idempotents, which proves the last statement without any boundedness or convergence assumption. $\square$

Corollary 2.2 (Localization and tame orbits). *Let H be tame and let $i \geq 1$. For every spectrum X with H -action, including spectra which are not p -local, the natural map gives*

$$L_{T(i)}(X_{hH}) \simeq (L_{T(i)}X)_{hH}.$$

Proof. Localization is a left adjoint into the local category, so

$$L_{T(i)}(X_{hH}) \simeq L_{T(i)}((L_{T(i)}X)_{hH}).$$

The localized object $L_{T(i)}X$ is p -local. By Lemma 2.1, its tame homotopy orbits are a finite-sum retract of it and are therefore still $T(i)$ -local. This removes the outer localization. $\square$

For a $T(n)$ -local commutative algebra A , its dualizable local modules form the category $\text{Mod}_A(\text{Sp}_{T(n)})^{\text{dbl}}$. For $n = 0$, every A -module is rational, so $\text{Mod}_A(\text{Sp}_{T(n)})^{\text{dbl}} = \text{Perf}(A)$. We use the following chromatic comparisons and vanishing results.

Proposition 2.3 (Chromatic inputs). *The following statements hold.*

- (i) *For an $\mathbb{E}_1$ -ring A and $i \geq 2$, the connective-cover map and cyclotomic trace give a natural zigzag of equivalences*

$$L_{T(i)}K(A) \longleftarrow L_{T(i)}K(\tau_{\geq 0}A) \longrightarrow L_{T(i)}(\text{TC}(\tau_{\geq 0}A)_p^\wedge). \quad (2.1)$$

This is [LMMT24, Corollary 4.30]. Derived p -completion does not change this localization: its fiber is p -inverted, hence $T(i)$ -acyclic for $i \geq 1$.

- (ii) *For $n \geq 1$ and A commutative, $L_{T(n)}A = 0$ implies $L_{T(n+1)}K(A) = 0$; see [CMNN, Theorem A].*
- (iii) *For $n \geq 1$ and a $T(n)$ -local commutative algebra A , $\text{Perf}(A) \rightarrow \text{Mod}_A(\text{Sp}_{T(n)})^{\text{dbl}}$ induces an equivalence after applying $L_{T(n+1)}K$. Moreover, $V \otimes M$ is an ordinary perfect A -module whenever V is a finite type- n spectrum and $M \in \text{Mod}_A(\text{Sp}_{T(n)})^{\text{dbl}}$. The comparison follows from [CMNN, Proposition 4.15 and Theorem A]; the assertion about $V \otimes M$ is proved in the proof of that proposition.*
- (iv) *For $n \geq 0$ and a $T(n)$ -local spectrum X with finite-group action, $L_{T(n)}(X^{tG}) = 0$. At $n = 0$, rational averaging gives $X^{tG} = 0$. For $n \geq 1$, this is telescopic Tate vanishing; see [Kuh04, Theorem 1.5, Proposition 1.10, and the proof of Corollary 1.7].*

The family-completeness input from [CMNN, proof of Theorem 6.5] will be stated precisely in Proposition 7.1. We also use the following categorical foundations. Nonconnective K -theory is a Morita-invariant localizing functor [BGT13, Theorem 9.8 and its consequence on localization]; it is lax symmetric monoidal [BGT15, Theorem 1.13]. Consequently, if a stable category $\mathcal{C}$ is A -linear, then $K(\mathcal{C})$ is a $K(A)$ -module. In particular, Proposition 2.3(ii) implies

$$n \geq 1, \quad L_{T(n)}A = 0 \quad \implies \quad L_{T(n+1)}K(\mathcal{C}) = 0 \quad (\mathcal{C} \text{ an } A\text{-linear category}). \quad (2.2)$$

Indeed, symmetric monoidal localization makes $L_{T(n+1)}K(\mathcal{C})$ a module over the zero algebra $L_{T(n+1)}K(A)$ in $\mathrm{Sp}_{T(n+1)}$.

Lemma 2.4. *For $n \geq 1$, every bounded-above spectrum is $T(n)$ -acyclic. In particular, $\tau_{\geq 0}A \rightarrow A$ is a $T(n)$ -equivalence for every ring spectrum A .*

Proof. Let $v : \Sigma^d V \rightarrow V$ be a v_n -self map of a finite type- n spectrum, with $d > 0$. A telescope representing $T(n)$ is $\mathrm{colim}_k \Sigma^{-kd} V$. If X is bounded above, then $V \otimes X$ is bounded above. For every fixed integer j ,

$$\pi_j(T(n) \otimes X) = \mathrm{colim}_k \pi_{j+kd}(V \otimes X) = 0.$$

The cofiber of $\tau_{\geq 0}A \rightarrow A$ is bounded above, proving the last assertion. $\square$

Lemma 2.5 (Local Galois structure). *For a $T(n)$ -local finite G -Galois extension $R \rightarrow S$, with $n \geq 0$, the following hold.*

- (i) S is dualizable and faithful as a local R -module.
- (ii) Local module descent induces an equivalence

$$\mathrm{Mod}_R(\mathrm{Sp}_{T(n)})^{\mathrm{dbl}} \xrightarrow{\simeq} (\mathrm{Mod}_S(\mathrm{Sp}_{T(n)})^{\mathrm{dbl}})^{hG}. \quad (2.3)$$

- (iii) For $H \leq G$, the extension $S^{hH} \rightarrow S$ is H -Galois. If H is normal, $R \rightarrow S^{hH}$ is G/H -Galois.

Proof. Dualizability is the finite Galois dualizability theorem, valid in the local symmetric monoidal category [Mat16, Proposition 6.14]. The local norm identifies $L_{T(n)}(S_{hG})$ with $S^{hG} = R$ by Proposition 2.3(iv). If $M \widehat{\otimes}_R S = 0$, tensoring this norm equivalence with M , and using preservation of colimits, gives $M = 0$. This proves faithfulness. The tensor functor with S is therefore conservative and, by dualizability, preserves limits. Descent follows from the comonadic descent argument for faithful Galois algebras; see [Mat16, §6.2, particularly Proposition 6.13]. Taking dualizable objects commutes with this limit, by [Lur17, Proposition 4.6.1.11]. This proves (2.3); precisely this local form is also recorded immediately after [CMNN, Proposition 4.15].

For completeness, the subgroup assertions can be checked after faithful base change along $R \rightarrow S$. Since S is dualizable, this base change commutes with homotopy fixed points, and (1.1) gives

$$S \widehat{\otimes}_R S^{hH} \simeq (S \widehat{\otimes}_R S)^{hH} \simeq \left(\prod_G S \right)^{hH} \simeq \prod_{G/H} S.$$

Here the action is the free permutation action on the indexing set; its homotopy limit is the displayed product. The asserted extensions become the split torsors on the corresponding finite sets. All their defining comparison maps are consequently equivalences after a conservative base change, hence before that base change. This also equips the normal-subgroup assertion with its coherent residual quotient action. $\square$

Proposition 2.6 (Equivariant comparison of module categories). *Let a finite group H act on a $T(n)$ -local commutative algebra S , with $n \geq 0$. Then the inclusion*

$$\mathrm{Perf}(S)^{hH} \longrightarrow (\mathrm{Mod}_S(\mathrm{Sp}_{T(n)})^{\mathrm{dbl}})^{hH} \quad (2.4)$$

induces an equivalence on $L_{T(n+1)}K$. In particular, if $R \rightarrow S$ is locally H -Galois, the canonical extension-of-scalars functor induces an equivalence

$$L_{T(n+1)}K(R) \xrightarrow{\simeq} L_{T(n+1)}K(\mathrm{Perf}(S)^{hH}). \quad (2.5)$$

Lemma 2.7 (Finite-local linearity). *Let $n \geq 1$, let $\mathcal{Q}$ be a small idempotent-complete stable p -local category, and let V be a finite type- n spectrum. If $V \otimes Q = 0$ for every $Q \in \mathcal{Q}$, then $\mathcal{Q}$ is linear over $L_{n-1}^{p,f}\mathbb{S}$, the finite smashing localization away from V .*

Proof. Pass to $\mathrm{Ind}(\mathcal{Q})$. Its canonical colimit-preserving spectral action annihilates V , and hence the localizing tensor ideal generated by V . By the universal property of localization it factors through the finite localization of spectra away from V . This localization is smashing, so its unit is the commutative idempotent algebra $L_{n-1}^{p,f}\mathbb{S}$ and the factorization is an $L_{n-1}^{p,f}\mathbb{S}$ -linear structure. It restricts to the compact objects: the unit acts as the identity, and each perfect module over that unit acts by finite colimits and retracts of the identity action. These preserve compact objects. The compact objects are $\mathcal{Q}$, giving the claim. Equivalently, all its mapping spectra are local for this finite localization. $\square$

Proof of Proposition 2.6. For $n = 0$, $\mathrm{Mod}_S(\mathrm{Sp}_{T(n)})^{\mathrm{dbl}} = \mathrm{Perf}(S)$, so the first assertion is an equivalence of categories; the second follows from Lemma 2.5. Assume henceforth that $n \geq 1$. The subcategory $\mathrm{Perf}(S)^{hH}$ is full, stable, and idempotent complete. Form the idempotent-completed Verdier quotient

$$\mathcal{Q} = ((\mathrm{Mod}_S(\mathrm{Sp}_{T(n)})^{\mathrm{dbl}})^{hH} / \mathrm{Perf}(S)^{hH})^{\mathrm{idem}}.$$

For a finite type- n spectrum V and $M \in (\mathrm{Mod}_S(\mathrm{Sp}_{T(n)})^{\mathrm{dbl}})^{hH}$, Proposition 2.3(iii) says that the underlying S -module of $V \otimes M$ is ordinary perfect. With its given action, it therefore belongs to $\mathrm{Perf}(S)^{hH}$. The action of V on the quotient is zero, also after idempotent completion. Lemma 2.7 makes $\mathcal{Q}$ linear over $L_{n-1}^{p,f}\mathbb{S}$. This algebra is $T(n)$ -acyclic, so (2.2) gives $L_{T(n+1)}K(\mathcal{Q}) = 0$. The localization sequence in nonconnective K -theory proves (2.4).

Now use the commuting square of extension-of-scalars and inclusion functors

$$\begin{array}{ccc} \mathrm{Perf}(R) & \xrightarrow{\quad} & \mathrm{Perf}(S)^{hH} \\ \downarrow & & \downarrow \\ \mathrm{Mod}_R(\mathrm{Sp}_{T(n)})^{\mathrm{dbl}} & \xrightarrow{\simeq} & (\mathrm{Mod}_S(\mathrm{Sp}_{T(n)})^{\mathrm{dbl}})^{hH}. \end{array}$$

After applying $L_{T(n+1)}K$, the left vertical arrow is an equivalence by Proposition 2.3(iii), the right one by the first part, and the bottom one by Lemma 2.5. The top arrow therefore induces (2.5). $\square$

Lemma 2.8. *A finite rational G -Galois extension $R \rightarrow S$ is faithful and descendable, and S is perfect over R . There is a natural symmetric monoidal equivalence*

$$\mathrm{Perf}(R) \simeq \mathrm{Perf}(S)^{hG}.$$

For every subgroup $H \leq G$, the map $S^{hH} \rightarrow S$ is an H -Galois extension with the same properties.

Proof. Rational averaging makes $R = S^{hG}$ a retract of S as an R -module. Thus the tensor ideal generated by S contains R , which is descendability. Dualizability from Lemma 2.5 is ordinary perfection in this case. The equivalence of perfect categories and the subgroup assertions are the rational case of the same lemma. $\square$

3. TWISTED HOCHSCHILD HOMOLOGY AND TAME COLIMITS

We identify twisted Hochschild homology for a generator of a tame cyclic group with the Hochschild homology of its commutative action colimit. The relative colimit formula will also be used to show that the auxiliary algebra in the Galois argument is chromatically acyclic.

3.1. Colimits of commutative algebras. For a space X and a commutative algebra A , form the colimit in $\mathbf{CAlg}$ and write

$$A \odot X := \operatorname{colim}_X \operatorname{const}(A).$$

This is a tensor with a space *in commutative algebras*; it is not the smash product of the underlying spectrum with $\Sigma_+^\infty X$.

Lemma 3.1 (Equivariant free resolutions). *Every object of $\operatorname{Fun}(BH, \mathbf{CAlg}(\operatorname{Sp}_{(p)}))$ has a functorial augmented simplicial resolution by algebras $\operatorname{Sym}(V_q)$, with $V_q \in \operatorname{Fun}(BH, \operatorname{Sp}_{(p)})$. Its geometric realization is the given algebra. Therefore a natural transformation between functors on this category that preserve geometric realizations is an equivalence if it is one on such free algebras.*

Proof. Let U be the forgetful functor. The free–forgetful adjunction gives the augmented bar resolution $(\operatorname{Sym}U)^{\bullet+1}B \rightarrow B$. The underlying augmented simplicial object is split by the unit of the adjunction. The forgetful functor preserves and detects geometric realizations, so the augmentation is a realization equivalence. The construction is functorial, hence also applies to coherent H -diagrams. For these standard facts about algebra objects and free resolutions, see [Lur17, Corollary 3.2.3.2, Proposition 4.7.3.14, and Remark 4.7.3.15]. $\square$

Lemma 3.2 (A constant tame diagram). *If H is tame and $A \in \mathbf{CAlg}(\operatorname{Sp}_{(p)})$, the augmentation*

$$A \odot BH \rightarrow A$$

is an equivalence. The same statement holds in $\mathbf{CAlg}(\operatorname{Sp}_{T(n)})$.

Proof. For $A = \operatorname{Sym}(V)$, preservation of colimits by Sym identifies this map with

$$\operatorname{Sym}(V \otimes \Sigma_+^\infty BH) \rightarrow \operatorname{Sym}(V).$$

The map inside Sym is an equivalence by Lemma 2.1 applied to the trivial action. Both sides preserve geometric realizations in A , so Lemma 3.1 proves the general case. Applying the symmetric monoidal localization proves the local version, since it preserves colimits of commutative algebras. $\square$

Lemma 3.3 (Absolute versus relative colimits). *Let R be a commutative algebra in a presentable symmetric monoidal category, and let B be a BH -diagram of commutative R -algebras. Let D be its colimit in $\mathbf{CAlg}$. Its colimit in $\mathbf{CAlg}_R$ is given by*

$$\operatorname{colim}_{BH} B \simeq R \otimes_{R \odot BH} D. \quad (3.1)$$

Consequently, for tame H in either setting of Lemma 3.2, the absolute colimit is also the relative R -algebra colimit. Its unit is an H -equivariant map to an algebra with trivial H -action in R -algebras.

Proof. An R -algebra cocone to the diagram consists of an absolute cocone together with agreement between its family of maps from R and the specified single map from R . Mapping the pushout on the right of (3.1) into an algebra E describes exactly these data: it is the homotopy fiber product of $\operatorname{Map}(D, E)$ and $\operatorname{Map}(R, E)$ over $\operatorname{Map}(R \odot BH, E)$. Taking the fiber over the specified R -algebra structure on E proves (3.1). The last assertion follows from Lemma 3.2 and the colimit adjunction. In particular, all the homotopies in the colimit cocone are relative to R . $\square$

3.2. Twisted circle colimits. Let $B \in \mathbf{CAlg}(\operatorname{Sp}_{(p)})$, and let g be an automorphism of B . Let B_g denote the B -bimodule whose left action is ordinary multiplication and whose right action is twisted by g . Write

$$\operatorname{THH}(B; B_g) = B \otimes_{B \otimes B} B_g. \quad (3.2)$$

In the last expression the two maps from $B \otimes B$ are multiplication and the twisted multiplication $(x, y) \mapsto xg(y)$. Since B is commutative, (3.2) can also be formed as a pushout in commutative algebras. Its underlying spectrum is the displayed derived relative tensor product. All tensors here

can be taken over $\mathbb{S}_{(p)}$: for p -local rings the absolute cyclic bar construction over $\mathbb{S}$ has the same underlying terms, because $\mathbb{S}_{(p)} \otimes \mathbb{S}_{(p)} \simeq \mathbb{S}_{(p)}$.

Lemma 3.4 (Twisted circle colimit). *Let $q : B\mathbb{Z} \rightarrow B\mathrm{Aut}(B)$ send 1 to g . Taking the colimit in $\mathrm{CAlg}(\mathrm{Sp}_{(p)})$, there is a natural equivalence*

$$\mathrm{THH}(B; B_g) \simeq \operatorname{colim}_{B\mathbb{Z}} q^* B. \quad (3.3)$$

For $g = 1$, this identifies $\mathrm{THH}(B)$ with $B \odot S^1$ as an underlying commutative algebra.

Proof. For a test commutative algebra E , put $Y = \mathrm{Map}_{\mathrm{CAlg}}(B, E)$. Mapping the pushout (3.2) into E gives the homotopy pullback of the diagonal $Y \rightarrow Y \times Y$ and the graph of $g^* : Y \rightarrow Y$. This is the homotopy equalizer of 1 and g^* , or equivalently the space of sections of the local system on a circle with monodromy g^* . That is exactly the mapping space from the colimit on the right of (3.3). Yoneda proves the claim. The same description, or the two-sided bar construction, identifies its underlying spectrum with the Hochschild tensor in (3.2). $\square$

Suppose now that $C_m = \langle g \rangle$ acts on B , with $(m, p) = 1$, and form the colimit in $\mathrm{CAlg}(\mathrm{Sp}_{(p)})$

$$C := \operatorname{colim}_{BC_m} B. \quad (3.4)$$

The colimit unit $u : B \rightarrow \mathrm{const}(C)$ is a coherent C_m -equivariant map. Applying the twisted-circle construction gives a specified natural map

$$u_g : \mathrm{THH}(B; B_g) \longrightarrow \mathrm{THH}(C). \quad (3.5)$$

The source has the residual C_m -action from the commuting actions on B , and the target has trivial C_m -action. More explicitly, this is obtained by pulling back the action along

$$C_m \times \mathbb{Z} \longrightarrow C_m, \quad (h, k) \longmapsto hg^k,$$

and taking the colimit in the $\mathbb{Z}$ -direction. Thus (3.5) is a map of C_m -objects, not just a choice of a nonequivariant map.

Proposition 3.5 (Tame twisted Hochschild comparison). *The map (3.5) is an equivalence of underlying spectra, and hence an equivalence in $\mathrm{Fun}(BC_m, \mathrm{Sp})$.*

Proof. Both sides, as functors of B with its C_m -action, preserve geometric realizations: by Lemma 3.4 they are composites of colimits of commutative algebras, and the forgetful functor preserves these realizations. It suffices by Lemma 3.1 to take $B = \mathrm{Sym}(V)$ with V a C_m -spectrum.

For this free algebra, the source and target of (3.5) are respectively

$$\mathrm{Sym}(V_{h\mathbb{Z}}) \quad \text{and} \quad \mathrm{Sym}(V_{hC_m} \otimes \Sigma_+^\infty S^1) = \mathrm{Sym}(V_{hC_m} \oplus \Sigma V_{hC_m}). \quad (3.6)$$

The actual map between them is induced by the colimit-unit map $V \rightarrow \mathrm{const}(V_{hC_m})$, after taking $\mathbb{Z}$ -orbits.

Apply the idempotent of Lemma 2.1:

$$V \simeq eV \oplus (1 - e)V, \quad e = \frac{1}{m} \sum_{j=0}^{m-1} g^j.$$

The first summand is coherently a trivial C_m -object and identifies with V_{hC_m} . On the second summand $1 - g$ is invertible, because the following identity holds in $\mathbb{Z}_{(p)}[C_m]$:

$$(1 - g) \left(\frac{1}{m} \sum_{j=0}^{m-1} (m - 1 - j) g^j \right) = 1 - e. \quad (3.7)$$

The homotopy orbits of a $\mathbb{Z}$ -action with generator g are $\mathrm{cofib}(1 - g)$. Thus

$$V_{h\mathbb{Z}} \simeq eV \oplus \Sigma eV.$$

On the augmentation summand, the colimit unit is an equivalence of C_m -objects: this follows either from the group-ring retract in Lemma 2.1, or from the unit–counit identity for the colimit adjunction applied to a trivial diagram. On the complementary summand, taking $\mathbb{Z}$ -orbits gives zero by (3.7). Hence it is the *specified* map in (3.6) that is an equivalence. Taking free algebras, and then the simplicial realization, proves the assertion. Conservativity of the forgetful functor from C_m -objects gives the equivariant assertion. $\square$

4. CYCLIC DESCENT AT POSITIVE HEIGHT

Throughout this section $n \geq 1$.

Proposition 4.1 (Tame cyclic descent). *For $n \geq 1$, Theorem 1.2 holds whenever G is cyclic of order prime to p .*

We first prove the assembly equivalence of Proposition 4.7 for a C_ℓ -Galois extension, where $\ell \neq p$ is prime. Its proof compares the cyclotomic assembly cofiber with that of a $T(n)$ -acyclic commutative algebra. Induction–restriction then turns assembly into Galois descent, and a tower of prime-order extensions handles every tame cyclic group.

We use the category CycSp of cyclotomic spectra in [NS18], with its exact, conservative, colimit-preserving forgetful functor $U : \text{CycSp} \rightarrow \text{Sp}$ [NS18, Corollary II.1.7]. The functor $X \mapsto \text{TC}(X)_p^\wedge$ is exact: TC is a mapping-spectrum functor [NS18, Definition II.1.8], and derived p -completion is exact. For a ring or a stable category, TC is applied to its topological Hochschild homology. The cyclotomic construction is functorial on stable categories and Morita invariant; this follows from [BM12, §4 and Theorems 5.11–5.12], together with the comparison in [NS18, §III.6]. Equivalently, it is stated directly in [Nik, Corollary 6.10]. The cyclotomic trace is natural for exact functors; see [BGT15, §7]. For the trace argument one can use connective K -theory throughout; its positive-height localization agrees with the nonconnective theory used here.

4.1. The auxiliary algebra.

Proposition 4.2. *Let $\ell \neq p$ be prime, and let $R \rightarrow S$ be a $T(n)$ -local C_ℓ -Galois extension, with $n \geq 1$. Form the colimit in $\text{CAlg}(\text{Sp}_{(p)})$*

$$C = \text{colim}_{BC_\ell} \tau_{\geq 0} S. \quad (4.1)$$

Then C is connective and $L_{T(n)}C = 0$.

Proof. Colimits of connective commutative ring spectra are connective; for example, this follows by resolving diagrams by free algebras on connective spectra. Thus C is connective. The connective-cover map $\tau_{\geq 0} S \rightarrow S$ is a $T(n)$ -equivalence by Lemma 2.4. Symmetric monoidal localization preserves colimits of commutative algebras. Taking the following colimit in $\text{CAlg}(\text{Sp}_{T(n)})$, we obtain

$$L_{T(n)}C \simeq \text{colim}_{BC_\ell} S.$$

By Lemmas 3.2 and 3.3, this is also the colimit in commutative R -algebras. Its unit $f : S \rightarrow L_{T(n)}C$ consequently satisfies

$$f \simeq f \circ g \quad \text{in } \text{CAlg}_R(\text{Sp}_{T(n)}) \quad (g \in C_\ell). \quad (4.2)$$

Choose $g \neq 1$. Under the Galois equivalence (1.1), the ordinary and g -twisted multiplications are the distinct coordinate projections

$$\mu_1, \mu_g : S \widehat{\otimes}_R S \simeq \prod_{h \in C_\ell} S \longrightarrow S.$$

Equation (4.2) gives a homotopy $f\mu_1 \simeq f\mu_g$: the two R -algebra maps are induced by the pairs (f, f) and (f, fg) . In the product algebra take the idempotent δ_1 which is 1 in the identity coordinate and 0 in the others. The two composites send this idempotent to $1_{L_{T(n)}C}$ and 0, respectively. Hence

$1_{L_{T(n)}C} = 0$ in $\pi_0(L_{T(n)}C)$. A unital ring spectrum with zero unit is contractible, since its identity map factors through multiplication by that unit. Thus $L_{T(n)}C = 0$. $\square$

4.2. Crossed products and Hochschild sectors. Let B be a p -local ring spectrum with a finite group action. Its crossed product $B \rtimes G$ is the $\mathbb{E}_1$ -ring with underlying spectrum $\bigoplus_{g \in G} Bu_g$ and multiplication

$$(bu_g)(cu_h) = b g(c) u_{gh}.$$

The crossed product is the algebra of the monad for equivariant B -modules. The displayed formula specifies the maps on summands; the monad retains the coherences of the action. We compute its derived tensor products by two-sided bar constructions.

Lemma 4.3 (Crossed-product modules). *There is a natural equivalence*

$$\mathrm{Mod}_{B \rtimes G} \simeq (\mathrm{Mod}_B)^{hG}.$$

If G is tame, it restricts to

$$\mathrm{Perf}(B \rtimes G) \simeq \mathrm{Perf}(B)^{hG}. \quad (4.3)$$

Proof. The forgetful functor from equivariant B -modules has a left adjoint given by the finite direct sum of the translates. It is conservative and creates the colimits needed in the monadicity theorem. Its monad is the crossed-product monad, so the first assertion follows; see the monadicity theorem [Lur17, Theorem 4.7.3.5].

An ordinary perfect $B \rtimes G$ -module is perfect after forgetting to B , since the free module $B \rtimes G$ is a finite direct sum of copies of B . Conversely, if an equivariant module M is perfect over B , its mapping spectrum in the equivariant category is

$$\mathrm{map}_{(\mathrm{Mod}_B)^{hG}}(M, N) = \mathrm{map}_{\mathrm{Mod}_B}(M, N)^{hG},$$

with the action furnished by the two equivariant structures. The first mapping functor preserves filtered colimits, and tame homotopy fixed points preserve them by Lemma 2.1. Thus M is compact in the equivariant module category, hence perfect over $B \rtimes G$. This proves (4.3). $\square$

The following sector formula also identifies the induction map.

Proposition 4.4 (The sector formula). *Let a finite group G act on a commutative ring spectrum B . There is a natural equivalence of underlying spectra*

$$U\mathrm{THH}(B \rtimes G) \simeq \left(\bigoplus_{g \in G} \mathrm{THH}(B; B_g) \right)_{hG}, \quad (4.4)$$

where G acts on the summands by conjugating their labels and acting on B . If G is abelian, this is

$$U\mathrm{THH}(B \rtimes G) \simeq \bigoplus_{g \in G} \mathrm{THH}(B; B_g)_{hG}. \quad (4.5)$$

The map $\mathrm{THH}(B)_{hG} \rightarrow \mathrm{THH}(B \rtimes G)$ induced by induction of perfect modules identifies, on underlying spectra, with the identity-labeled summand.

Proof. Write $B^e = B \otimes B^{\mathrm{op}}$, and use the same enveloping-algebra notation for $B \rtimes G$. The diagonal action gives a subalgebra

$$B^e \rtimes G \longrightarrow (B \rtimes G)^e$$

which sends the group symbol to

$$d_g = u_g \otimes (u_g^{-1})^{\mathrm{op}}.$$

Consider B as a right $(B^e \rtimes G)$ -module, with

$$b \cdot (x \otimes y^{\mathrm{op}}) = ybx, \quad b \cdot d_g = g^{-1}(b).$$

Let $(B \rtimes G)_r$ be the right $(B \rtimes G)^e$ -module with $z \cdot (a \otimes c^{\text{op}}) = cza$. Multiplication gives a right $(B \rtimes G)^e$ -linear equivalence

$$B \otimes_{B^e \rtimes G} (B \rtimes G)^e \xrightarrow{\simeq} (B \rtimes G)_r, \quad b \otimes (a \otimes c^{\text{op}}) \mapsto cba. \quad (4.6)$$

Here is a direct verification of the equivalence. As a left $(B^e \rtimes G)$ -module, $(B \rtimes G)^e$ is a direct sum of copies of $(B^e \rtimes G)$ represented by $u_t \otimes 1$, $t \in G$. On group labels this is the factorization

$$u_r \otimes u_s^{\text{op}} = d_{s^{-1}}(u_{sr} \otimes 1).$$

The coefficient maps are invertible translates of B^e . Tensoring with B , the summand t in (4.6) maps by $b \mapsto bu_t$ onto the t -summand of $(B \rtimes G)$. Thus the map is an equivalence. Its compatibility with the $(B^e \rtimes G)$ -relations follows, in particular, from

$$(cu_g^{-1})b(u_g a) = c g^{-1}(b)a.$$

Using the usual enveloping-algebra model for Hochschild homology, (4.6) gives

$$U\text{THH}(B \rtimes G) = (B \rtimes G)_r \otimes_{(B \rtimes G)^e} (B \rtimes G) \simeq B \otimes_{B^e \rtimes G} (B \rtimes G). \quad (4.7)$$

For a right $(B^e \rtimes G)$ -module N and a left $(B^e \rtimes G)$ -module M , the relative bar construction proves

$$N \otimes_{B^e \rtimes G} M \simeq (N \otimes_{B^e} M)_{hG}. \quad (4.8)$$

Indeed, the relative two-sided bar in degree q is

$$N \otimes_{B^e} \underbrace{(B^e \rtimes G) \otimes_{B^e} \cdots \otimes_{B^e} (B^e \rtimes G)}_{q \text{ factors}} \otimes_{B^e} M.$$

Expand $(B^e \rtimes G)$ as the direct sum of its group-labeled copies of B^e . Moving the group symbols across the tensor signs identifies this with a sum indexed by G^q of $N \otimes_{B^e} M$. The inner face maps multiply adjacent symbols, the end faces use the two module actions, and degeneracies insert the identity. These are exactly the group bar maps for the action

$$g(n \otimes m) = n d_g^{-1} \otimes d_g m.$$

Realization proves (4.8).

Apply it with $N = B$ and $M = (B \rtimes G)$. The left B^e -module $(B \rtimes G)$ decomposes as $\bigoplus_g B_g$: on zu_g the action of $x \otimes y^{\text{op}}$ is $(xz g(y))u_g$. Thus

$$B \otimes_{B^e} (B \rtimes G) \simeq \bigoplus_g \text{THH}(B; B_g).$$

The action in (4.8) acts on B by g and on $(B \rtimes G)$ by conjugation by u_g . It therefore has exactly the label-conjugation action asserted in (4.4). Equations (4.7) and (4.8) prove the formula, and for abelian G yield (4.5).

It remains to identify the map. Before taking group orbits, the identity-sector inclusion is

$$B \otimes_{B^e} B \longrightarrow B \otimes_{B^e} (B \rtimes G),$$

induced by the identity summand $B = B_1 \subset (B \rtimes G)$. Under (4.6), its composite into $(B \rtimes G)_r \otimes_{(B \rtimes G)^e} (B \rtimes G)$ is the usual Hochschild map of the algebra inclusion $B \rightarrow (B \rtimes G)$. The group-bar coherences are precisely conjugation by the units u_g , which also give the equivariant cocone for the induction functor $\text{Perf}(B) \rightarrow \text{Perf}(B \rtimes G)$. Consequently the map after group orbits is the assembly map of that induction functor. Every construction above is functorial in coherent equivariant algebra maps, proving the required naturality as well. $\square$

4.3. Cyclotomic assembly. For a tame finite group G acting on B , induction of perfect modules gives a coherent cocone from $\text{Perf}(B)$ to $\text{Perf}(B \rtimes G)$. For left modules, put $g_*M = B \otimes_{B,g} M$. The natural equivalence

$$\text{Ind}(g_*M) \simeq (B \rtimes G) \otimes_{B,g} M \longrightarrow (B \rtimes G) \otimes_B M, \quad a \otimes m \longmapsto au_g \otimes m$$

is balanced because $g(b)u_g = u_gb$. Its cocycle identity is $u_gu_h = u_{gh}$. Applying the cyclotomic Hochschild functor therefore produces a natural map in CycSp

$$\theta_B : \text{THH}(B)_{hG} \longrightarrow \text{THH}(B \rtimes G). \quad (4.9)$$

In this expression the homotopy orbits are formed in CycSp . Its underlying map is the assembly map in Proposition 4.4. If G is abelian, exactness and colimit preservation of U give

$$U\text{cofib}_{\text{CycSp}}(\theta_B) \simeq \bigoplus_{g \neq 1} \text{THH}(B; B_g)_{hG}. \quad (4.10)$$

We will use this underlying-sector description to detect equivalences of cyclotomic cofibers.

Lemma 4.5 (Comparison of the cyclotomic defects). *Let $\ell \neq p$ be prime, and let B be a commutative p -local ring spectrum with C_ℓ -action. Form $C = \text{colim}_{BC_\ell} B$ in $\text{CAlg}(\text{Sp}_{(p)})$ and give it the trivial C_ℓ -action. The colimit unit induces an equivalence in CycSp :*

$$\text{cofib}_{\text{CycSp}}(\theta_B) \xrightarrow{\simeq} \text{cofib}_{\text{CycSp}}(\theta_C). \quad (4.11)$$

Proof. The coherent equivariant unit $B \rightarrow \text{const}(C)$ induces an actual crossed-product algebra map

$$B \rtimes C_\ell \longrightarrow C[C_\ell].$$

It is functorial for induction, giving a commutative square in CycSp

$$\begin{array}{ccc} \text{THH}(B)_{hC_\ell} & \xrightarrow{\theta_B} & \text{THH}(B \rtimes C_\ell) \\ \downarrow & & \downarrow \\ \text{THH}(C)_{hC_\ell} & \xrightarrow{\theta_C} & \text{THH}(C[C_\ell]). \end{array} \quad (4.12)$$

Taking cofibers constructs the map in (4.11). For each $g \neq 1$, the element g generates C_ℓ . The map on the g -sector in (4.10) is the map $\text{THH}(B; B_g)_{hC_\ell} \rightarrow \text{THH}(C)_{hC_\ell}$ of Proposition 3.5, hence an equivalence. The map in (4.11) is therefore an equivalence on underlying spectra. It is already a morphism of cyclotomic spectra by (4.12); conservativity of U makes it an equivalence in CycSp . $\square$

Remark 4.6. The crossed-product map constructs the morphism (4.11) in CycSp . Thus its Frobenius compatibility follows from functoriality. The sector computation is used to detect that this morphism is an equivalence.

Proposition 4.7 (Assembly equivalence). *Let $R \rightarrow S$ be a $T(n)$ -local C_ℓ -Galois extension, with $n \geq 1$ and $\ell \neq p$ prime. Then the natural induction maps*

$$\begin{aligned} (L_{T(n+1)}K(\tau_{\geq 0}S))_{hC_\ell} &\xrightarrow{\simeq} L_{T(n+1)}K((\tau_{\geq 0}S) \rtimes C_\ell), \\ (L_{T(n+1)}K(S))_{hC_\ell} &\xrightarrow{\simeq} L_{T(n+1)}K(S \rtimes C_\ell) \end{aligned} \quad (4.13)$$

are equivalences.

Proof. Use the algebra C of Proposition 4.2, and apply Lemma 4.5 to $\tau_{\geq 0}S$. All underlying Hochschild spectra in (4.12) are p -local, so multiplication by ℓ is invertible on them. It is therefore also invertible on their cyclotomic objects, by conservativity of U . Lemma 2.1, applied to the exact functor $\text{TC}(-)_p^\wedge$, gives

$$\text{TC}(\text{THH}(\tau_{\geq 0}S)_{hC_\ell})_p^\wedge \simeq (\text{TC}(\tau_{\geq 0}S)_p^\wedge)_{hC_\ell},$$

and likewise for C . Apply $\mathrm{TC}(-)_p^\wedge$ to (4.11). Exactness identifies the result with an equivalence

$$\begin{aligned} \mathrm{cofib}((\mathrm{TC}(\tau_{\geq 0}S)_p^\wedge)_{hC_\ell} &\longrightarrow \mathrm{TC}(\tau_{\geq 0}S \rtimes C_\ell)_p^\wedge) \\ &\simeq \mathrm{cofib}((\mathrm{TC}(C)_p^\wedge)_{hC_\ell} \longrightarrow \mathrm{TC}(C[C_\ell])_p^\wedge). \end{aligned} \quad (4.14)$$

Both C and $C[C_\ell]$ are connective commutative algebras. Commutativity of $C[C_\ell]$ uses that C_ℓ is abelian. Proposition 4.2 gives $L_{T(n)}C = 0$, and its finite direct-sum description gives $L_{T(n)}C[C_\ell] = 0$. Chromatic vanishing and (2.1), at height $n+1 \geq 2$, imply

$$L_{T(n+1)}(\mathrm{TC}(C)_p^\wedge) = 0, \quad L_{T(n+1)}(\mathrm{TC}(C[C_\ell])_p^\wedge) = 0.$$

Hence (4.14) vanishes after $L_{T(n+1)}$. Naturality of the cyclotomic trace for induction identifies this with the cofiber of the first map in (4.13). This proves that map is an equivalence. Corollary 2.2 identifies the localized K -theory orbits with the orbits of $L_{T(n+1)}K(\tau_{\geq 0}S)$.

The crossed product is a finite sum of translates on underlying spectra, and connective cover is compatible with its multiplication and action. Thus

$$\tau_{\geq 0}(S \rtimes C_\ell) \simeq (\tau_{\geq 0}S) \rtimes C_\ell.$$

The natural connective-cover equivalences in (2.1) identify the assembly maps for $\tau_{\geq 0}S$ and S , proving the second assertion. $\square$

4.4. Cyclic descent.

Proof of Proposition 4.1. First consider C_ℓ , with $\ell \neq p$ prime. Lemma 4.3 identifies $\mathrm{Perf}(S \rtimes C_\ell)$ with $\mathrm{Perf}(S)^{hC_\ell}$. Induction and restriction give maps

$$(L_{T(n+1)}K(S))_{hC_\ell} \xrightarrow{a} L_{T(n+1)}K(\mathrm{Perf}(S)^{hC_\ell}) \xrightarrow{r} (L_{T(n+1)}K(S))^{hC_\ell}.$$

The first is an equivalence by Proposition 4.7. The composite is the norm. Indeed, the restriction of induction is the direct sum of the translates, with its usual coherent permutation structure; its effect is the sum of the action maps. This is also the standard induction–restriction description of the K -theoretic norm in [CMNN, §2, equation (2.1) and the subsequent discussion]. The norm is an equivalence by Lemma 2.1; therefore r is an equivalence. Finally, (2.5) identifies $L_{T(n+1)}K(R) \rightarrow L_{T(n+1)}K(\mathrm{Perf}(S)^{hC_\ell})$ as an equivalence. Composing it with r is exactly extension of scalars to S with its canonical descent data. This proves the canonical comparison (1.2) for prime-order C_ℓ .

Now let G be any cyclic group of order prime to p . Induct on its order. Choose a normal subgroup H of prime order. Lemma 2.5 supplies the local Galois extensions $S^{hH} \rightarrow S$ and $R \rightarrow S^{hH}$ with groups H and G/H . The comparison for the first extension is natural under its residual G/H -action. By the prime-order case, the induction hypothesis, and iterated homotopy fixed points,

$$L_{T(n+1)}K(R) \simeq (L_{T(n+1)}K(S^{hH}))^{h(G/H)} \simeq ((L_{T(n+1)}K(S))^{hH})^{h(G/H)} \simeq (L_{T(n+1)}K(S))^{hG}.$$

Every arrow is a canonical comparison, proving the assertion in the required natural form. $\square$

5. TORSION LINES OVER RATIONAL RINGS

The trace comparison used at positive input height has output height at least two. At output height one it already fails on rational fields.

Proposition 5.1. *For every rational ring spectrum A , $\mathrm{TC}(A)_p^\wedge = 0$. In contrast, $L_{T(1)}K(H\mathbb{C})$ is nonzero. Thus the trace equivalence (2.1) is false at output height 1, even for a discrete rational field.*

Proof. The underlying Hochschild spectrum of a rational ring is rational: every term in its cyclic bar is rational, as is its realization. Multiplication by p is therefore an equivalence on the underlying spectrum of $\mathrm{THH}(A)$, hence also on its cyclotomic object by conservativity. Exactness of TC preserves this scalar equivalence. Its derived p -completion is consequently zero.

For $A = H\mathbb{C}$, Suslin's comparison with topological K -theory gives

$$L_{T(1)}K(H\mathbb{C}) \simeq KU_p^\wedge \neq 0;$$

this standard consequence is recalled and used in [CMNN, Introduction, p. 2]. It contradicts a putative height-1 version of (2.1). $\square$

Also, for a nonconnective rational S , the map $\tau_{\geq 0}S \rightarrow S$ need not be a rational equivalence. These two facts explain the separate argument of Sections 5 and 6.

An invertible A -module L is called *even* if its self-braiding $L \otimes_A L \rightarrow L \otimes_A L$ is the identity. For rational modules, this implies

$$\mathrm{Sym}_A^j(L) \simeq L^{\otimes j} \quad (j \geq 0), \quad \mathrm{Sym}_A^j(\Sigma L) = 0 \quad (j \geq 2). \quad (5.1)$$

Indeed, finite-group coinvariants are exact rationally, and the permutation actions here are respectively trivial and sign actions. Evenness is preserved by base change, duals, and even suspensions.

We use rational commutative differential graded algebras when giving strict models. Their grading is cohomological: d raises degree by one and $H^i(A) = \pi_{-i}(A)$. Elsewhere Σ and π_n use homotopy grading.

Theorem 5.2. *Let R be a rational commutative ring spectrum, and let L be an even invertible R -module with $L^{\otimes m} \simeq R$, where $m \geq 1$ and $(m, p) = 1$. Then*

$$[L] = 1 \quad \text{in } \pi_0 L_{T(1)}K(R).$$

More generally, tensoring by L acts as the identity on every $\mathbb{A}^1$ -invariant localizing invariant with values in an idempotent-complete stable category on which multiplication by m is invertible.

The proof will make the complementary averaging summand for L a retract of zero. Lemma 5.10 supplies a finite sequence of section nullifications ending at the zero algebra. Corollary 5.9 handles odd sections by $\mathbb{A}^1$ -invariance, while the Euler transfer for even sections is invertible on that summand. We first establish the invariance assertion.

5.1. Rational $\mathbb{A}^1$ -invariance.

Lemma 5.3. *The functor $L_{T(1)}K$ on rational linear stable categories is Morita invariant, localizing, and $\mathbb{A}^1$ -invariant. It has the usual K -theory action of perfect modules, and satisfies the projection formula for restriction of scalars along an algebra which is perfect as a module over the source.*

Proof. The localizing and multiplicative properties follow from [BGT13, Theorem 9.8] and [BGT15, Theorem 1.13], retained by exact symmetric monoidal localization. Tabuada [Tab, Theorem 1.2(i)] proves $\mathbb{A}^1$ -invariance of $K(-; \mathbb{Z}/p)$ for every rational dg category. Thus the cofiber of $K(\mathcal{C}) \rightarrow K(\mathcal{C}[t])$ has multiplication by p invertible, and is $T(1)$ -acyclic. This proves the stated invariance. The projection formula is induced by the equivalence of exact functors

$$\mathrm{res}((P \otimes_A B) \otimes_B Q) \simeq P \otimes_A \mathrm{res}(Q).$$

$\square$

5.2. A natural splitting principle.

Lemma 5.4. *Let $\mathcal{G}$ be an endofunctor of a category $\mathcal{D}$ and let $h : \text{id} \rightarrow \mathcal{G}$ be a natural transformation. Let $F : \mathcal{D} \rightarrow \mathcal{V}$ take values in a stable category. Suppose that there is a natural transformation $r : F\mathcal{G} \rightarrow F$ with $r_A F(h_A) = \text{id}$, and that*

$$F(\mathcal{G}(h_A)) \simeq F(h_{\mathcal{G}(A)})$$

for every A . Then $F(h_A)$ is an equivalence with inverse r_A .

Proof. Naturality of r applied to h_A gives

$$F(h_A)r_A = r_{\mathcal{G}(A)}F(\mathcal{G}(h_A)) = r_{\mathcal{G}(A)}F(h_{\mathcal{G}(A)}) = \text{id}.$$

The other composite is the assumed identity. $\square$

We will construct r from a natural nullhomotopy in a localization sequence. A polynomial homotopy will identify the two maps in the hypothesis.

5.3. Killing a closed odd element. In the next two subsections, F denotes any Morita-invariant, localizing, $\mathbb{A}^1$ -invariant functor on rational dg categories, with values in an idempotent-complete stable category. In particular, one may take $F = L_{T(1)}K$ or $F = K(-; \mathbb{Z}/p^a)$, with $a \geq 1$.

Proposition 5.5. *Let A be an arbitrary rational commutative dg algebra and let $\theta \in Z^{2r+1}(A)$, with $r \in \mathbb{Z}$. The canonical map*

$$F(A) \longrightarrow F(A[x], |x| = 2r, dx = \theta)$$

is an equivalence. This construction is natural in (A, θ) .

Proof. Work in commutative $\Lambda_{\mathbb{Q}}(\theta)$ -algebras, where the exterior algebra has zero differential. Set

$$D(A) = A \otimes_{\Lambda_{\mathbb{Q}}(\theta)} \mathbb{Q} \simeq (A[x], dx = \theta).$$

The displayed model is valid in all degrees: $\Lambda_{\mathbb{Q}}(\theta)[x]$, with $dx = \theta$, is a semifree resolution of $\mathbb{Q}$ over $\Lambda_{\mathbb{Q}}(\theta)$.

Consider the A -linear dg category $\mathcal{T}_A$ on two objects E_-, E_+ , with endomorphism algebras A , no morphisms from E_+ to E_- , and

$$\mathcal{T}_A(E_-, E_+) = A\{s, t\}, \quad |s| = 0, \quad |t| = 2r, \quad ds = 0, \quad dt = \theta s.$$

Take its pretriangulated idempotent completion and let $K_s = \text{cofib}(s : E_- \rightarrow E_+)$. Inverting s gives a compactly generated localization with kernel $\text{thick}(K_s)$. A module over the localized category consists of a single A -module with an operator x of degree $2r$ satisfying $dx = \theta$. Hence the quotient is Morita equivalent to $D(A)$. This uses that the free associative algebra on one *even* generator is the polynomial algebra on that generator.

Here is a strict model for the support algebra. In $\text{End}(K_s)$ let c be the projection onto the shifted source summand. With a consistent cone convention, the elements s, t in this endomorphism algebra have degrees $1, 2r + 1$, and

$$dc = s, \quad dt = \theta s, \quad c^2 = c, \quad tc = t, \quad ct = 0, \quad t^2 = 0.$$

The element $\epsilon = t + \theta c$ is closed and satisfies $\epsilon^2 = -\theta\epsilon$. The strict algebra map

$$C_A = A \oplus A\epsilon \longrightarrow \text{End}(K_s), \quad |\epsilon| = 2r + 1, \quad d\epsilon = 0, \quad \epsilon^2 = -\theta\epsilon \quad (5.2)$$

is a quasi-isomorphism. Indeed, after this change of basis its complement as an A -module is the contractible pair c, s . The usual graded commutation rule holds between ϵ and A . We obtain an exact sequence of stable categories

$$\text{Perf}(C_A) \longrightarrow \mathcal{T}_A \longrightarrow \text{Perf}(D(A)). \quad (5.3)$$

There are two strict A -linear coordinate functors $q_-, q_+ : \mathcal{T}_A \rightarrow \text{Perf}(A)$: each keeps its indicated vertex and sends the other vertex and both arrows to zero. Additivity gives

$$(F(q_-), F(q_+)) : F(\mathcal{T}_A) \simeq F(A) \oplus F(A).$$

Moreover $q_+(K_s) = A$ and $q_-(K_s) = \Sigma A$. On the former, c, t act as zero; on the latter, c acts as the identity and t as zero. Thus, in these coordinates, the first map in (5.3) is

$$(-F(p_1), F(p_0)) : F(C_A) \longrightarrow F(A)^2, \quad p_0(\epsilon) = 0, \quad p_1(\epsilon) = \theta. \quad (5.4)$$

The minus sign records the suspension.

Because θ is odd and A is rational, $\theta^2 = 0$. Hence

$$C_A \longrightarrow A[z], \quad \epsilon \longmapsto z\theta, \quad |z| = 0, \quad dz = 0$$

is a strict algebra homotopy from p_0 to p_1 , natural in (A, θ) . The sum map $F(A)^2 \rightarrow F(A)$ therefore has a natural nullhomotopy after composition with (5.4). Passing to the cofiber in (5.3) constructs a natural map

$$r_A : F(D(A)) \longrightarrow F(A).$$

The functor $\text{Perf}(A) \rightarrow \mathcal{T}_A$ at E_+ has coordinates $(0, \text{id})$, and its image in the quotient is extension along $h_A : A \rightarrow D(A)$. It follows that $r_A F(h_A) = \text{id}$.

To apply Lemma 5.4, write

$$D(D(A)) = A[x_1, x_2], \quad dx_1 = dx_2 = \theta.$$

The maps $D(h_A)$ and $h_{D(A)}$ send x to x_2 and x_1 . They are joined by the polynomial homotopy

$$x \longmapsto x_1 + z(x_2 - x_1).$$

It commutes with the differential and is an algebra map because x is even. Thus their images under F agree. The natural splitting lemma proves the proposition. $\square$

Corollary 5.6. *For the associative algebra C_A of (5.2), the augmentation $C_A \rightarrow A$, $\epsilon \mapsto 0$, induces an equivalence on F .*

Proof. In the localization sequence, the first map is homotopic to $(-F(p_0), F(p_0))$. The second map has cofiber target $F(D(A))$, and its restriction to the second summand is the equivalence just proved. An invertible change of coordinates identifies its fiber with $F(A)$ and the resulting fiber map with $F(p_0)$. $\square$

Remark 5.7. Evenness of the affine coordinate is needed here. Killing an even element would give a free associative *odd* generator in the localized Kronecker category. Its square need not vanish, so that localization does not model the commutative quotient.

5.4. Affine torsors for even invertible modules.

Theorem 5.8. *Let A be a rational commutative ring spectrum, let J be an even invertible A -module, and let $\alpha : J \rightarrow \Sigma A$. Form the cofiber sequence*

$$A \xrightarrow{s} V \longrightarrow J \xrightarrow{\alpha} \Sigma A$$

and the commutative algebra

$$D = \text{Sym}_A(V) \otimes_{\text{Sym}_A(A)} A, \quad (5.5)$$

where the two maps from $\text{Sym}_A(A)$ are induced by s and the unit $1 : A \rightarrow A$. Then $F(A) \rightarrow F(D)$ is an equivalence, naturally in the data and their base changes.

Proof. Use the A -linear two-object category with forward morphism module V , reverse morphism module zero, and endomorphism algebras A . The arrow s defines $K_s = \text{cofib}(s)$. There are the same two coordinate functors q_-, q_+ as above, and localization at s has kernel generated by K_s . We verify both the quotient algebra and the requisite homotopy of support actions before applying F .

Construct the commutative, strongly $\mathbb{Z}$ -graded frame algebra

$$P = \bigoplus_{a \in \mathbb{Z}} J^{\otimes a}.$$

The multiplication $P_a \otimes_A P_b \simeq P_{a+b}$ uses invertibility; its commutative structure uses evenness and (5.1). Equivalently, form $\text{Sym}_A(J \oplus J^\vee)$ and invert the pairing by setting it equal to 1. There is an equivalence between A -modules and graded P -modules, given by $M \mapsto P \otimes_A M$ and its inverse taking weight zero. To check it, use the invertibility of P_a to identify the weight- a part of any graded P -module with P_a tensored with its weight-zero part. These functors are symmetric monoidal. In particular, they also identify algebras, modules over algebras, and their coherent homotopies. Forgetting weights, $P \otimes_A -$ is conservative, since A is its weight-zero summand.

Over P the module J has a canonical homogeneous frame. In that frame the extension defining V has the strict model $P\{s, t\}$ with $|s| = |t| = 0$, $ds = 0$, $dt = \theta s$, for a homogeneous closed element θ of cohomological degree 1. Both t and θ have the same weight. The scalar construction of Proposition 5.5 therefore applies. Localization at s is the polynomial algebra $P[x]$, $dx = \theta$, which is also the base change of (5.5).

More intrinsically, the quotient of the two-object category is an associative algebra of affine coordinates, with a canonical map to D . The preceding computation makes this map an equivalence after tensoring with P , so conservativity makes it an equivalence over A . One can also check this with its polynomial filtration: its associated graded is the tensor algebra on J , whose degree- a piece agrees with $\text{Sym}_A^a(J)$ by evenness.

In the frame, the support algebra and the two strict coordinate actions are exactly (5.2) and (5.4). The homotopy $\epsilon \mapsto z\theta$ is homogeneous, with z of weight zero. It is consequently a map of graded algebras, and descends, together with its endpoint identifications, to A . Thus the homotopy is constructed over A before applying F .

To record its naturality, use a single universal strict model. Take the graded dg algebra $\Lambda_{\mathbb{Q}}(\theta)$, where θ has cohomological degree 1 and frame weight w . In its Kronecker model, s, c have weight zero and t, θ, ϵ have weight w , while z has weight zero. The fixed strict graded map $C_{\Lambda_{\mathbb{Q}}(\theta)} \rightarrow \Lambda_{\mathbb{Q}}(\theta)[z]$, $\epsilon \mapsto z\theta$, is base changed along the graded classifying map $\Lambda_{\mathbb{Q}}(\theta) \rightarrow P$ for the extension and then descended by taking weight zero. The classifying map is understood in the derived category of graded algebras; strict representatives present this construction without becoming additional choices in its definition. Since the frame algebra commutes with every base change $A \rightarrow A'$, all these maps and their homotopies are natural under base change. In particular the resulting retraction below has the naturality required in Lemma 5.4.

The sum of the two coordinate maps thus has a natural nullhomotopy on the support category. As in the scalar case, the localization sequence gives a natural retraction

$$r : F(D) \longrightarrow F(A), \quad rF(A \rightarrow D) = \text{id}.$$

Finally, the two maps $D \rightrightarrows D \otimes_A D$ have a canonical affine interpolation. In a frame it is $x \mapsto x_1 + z(x_2 - x_1)$; intrinsically it is the affine linear combination of the two maps $V \rightarrow D \otimes_A D$ that both send s to 1. It defines an algebra map from (5.5), and commutes with base change. Apply Lemma 5.4 to the endofunctor of A -algebras given by tensoring with D . The retraction is also a right inverse. $\square$

Corollary 5.9. *Let L be an even invertible A -module, let $q \in \mathbb{Z}$ be odd, and let $g : \Sigma^q L^\vee \rightarrow A$. Then the commutative nullification*

$$A \longrightarrow A \otimes_{\mathrm{Sym}_A(\Sigma^q L^\vee)} A$$

induces an equivalence on F ; the second map from the symmetric algebra is the zero augmentation.

Proof. Put $J = \Sigma^{q+1} L^\vee$ and $\alpha = \Sigma g$. Then J is even. Both the nullification in the statement and (5.5) corepresent a nullhomotopy of g after base change: a map out of the latter is a retraction of s and hence a nullhomotopy of its boundary. They are thus canonically equivalent. Now apply Theorem 5.8. $\square$

5.5. Finite annihilation by sections of a line.

Lemma 5.10. *Let R be any commutative ring spectrum and let L be an invertible R -module. There is a finite sequence*

$$R = A_0 \longrightarrow A_1 \longrightarrow \cdots \longrightarrow A_N = 0$$

such that each step has the form

$$A_{i+1} = A_i \otimes_{\mathrm{Sym}_R(\Sigma^{q_i} L^\vee)} R, \tag{5.6}$$

where $q_i \in \mathbb{Z}$, the first structure map is induced by an R -linear map $\Sigma^{q_i} L^\vee \rightarrow A_i$ and the second is the zero augmentation.

Proof. Since L is perfect, it is a retract of a finite cell R -module Q . Thus $P = L^\vee \otimes_R Q$ is a finite cell module with cells $\Sigma^q L^\vee$ for $q \in \mathbb{Z}$, and $R = L^\vee \otimes_R L$ is a retract of P .

We first prove that any map $f : P' \rightarrow A$ from a finite $L^\vee$ -cell module to a commutative R -algebra becomes null after finitely many steps of the form (5.6). Induct on the number of cells. For a last-cell sequence

$$P'' \longrightarrow P' \longrightarrow \Sigma^q L^\vee,$$

first use induction to make the restriction $f|_{P''}$ null over an algebra A' . Choose this nullhomotopy. Exactness of mapping spaces gives a factorization of $P' \rightarrow A'$ through a map $\Sigma^q L^\vee \rightarrow A'$. One further pushout $A' \otimes_{\mathrm{Sym}_R(\Sigma^q L^\vee)} R$ kills this factor, and therefore kills f . This completes the induction. Every attachment is a single allowed cell, and there are at most as many as there are cells in P' .

Apply the assertion to the retraction $r : P \rightarrow R$. After a finite sequence, its composite $P \rightarrow A_N$ is null. Precomposing with the section $R \rightarrow P$ makes the unit $R \rightarrow A_N$ null. Hence $1 = 0$ in $\pi_0 A_N$, so the unital algebra A_N is zero. $\square$

The map $\Sigma^q L^\vee \rightarrow A_i$ is, by duality, a homogeneous class in $\pi_q(L \otimes_R A_i)$. Thus the lemma uses only the successive killing of sections of the transported line.

Proof of Theorem 5.2. We first apply the argument to $L_{T(1)}K$. The class $[L]$ in $\pi_0 L_{T(1)}K(R)$ satisfies $[L]^m = 1$. Since $L_{T(1)}K(R)$ is p -local, m is invertible. Define the complementary idempotents

$$e = \frac{1}{m} \sum_{j=0}^{m-1} [L]^j, \quad \epsilon = 1 - e.$$

For every R -algebra A , put $L_A = A \otimes_R L$ and let e_A, ϵ_A be the images of e, ϵ . We show that the summand $(L_{T(1)}K(A))_{\epsilon_A}$ maps split injectively along each step of Lemma 5.10.

Define the Laurent polynomial

$$b(z) = \frac{1}{m} \sum_{j=0}^{m-1} (m-1-j)z^{-j}.$$

Telescoping the finite sum and using $[L]^m = 1$ gives

$$(1 - [L]^{-1})b([L]) = \epsilon. \tag{5.7}$$

Consequently $1 - [L]^{-1}$ acts invertibly on $(L_{T(1)}K(R))_\epsilon$, with inverse $b([L])$. The assertion remains true after every base change.

Apply Lemma 5.10 to L . Consider one step, over a current algebra A , killing $g : E = \Sigma^q L_A^\vee \rightarrow A$, with $q \in \mathbb{Z}$. If q is even, E is an even invertible module. Its one-line Koszul resolution gives

$$C = A \otimes_{\text{Sym}_A(E)} A \simeq \text{cofib}(E \xrightarrow{g} A) \quad \text{as an } A\text{-module.} \quad (5.8)$$

For example, the Koszul algebra has underlying module $A \oplus \Sigma E$, with differential determined by g ; its higher symmetric powers vanish by (5.1). This argument works for an arbitrary even line by its frame algebra, or directly from the cofiber sequence

$$\text{Sym}_A(E) \otimes_A E \longrightarrow \text{Sym}_A(E) \longrightarrow A.$$

Indeed, the first map identifies each tensor power $E^{\otimes j}$ with the summand of polynomial degree $j \geq 1$, so its cofiber is exactly the degree-zero summand. Tensoring this resolution along the map classified by g proves (5.8) without any regularity or flatness assumption on homotopy groups. Thus C is perfect over A and

$$[C] = 1 - [E] = 1 - [L_A]^{-1} \quad \text{in } K_0(A).$$

Restriction of scalars gives a transfer $t : L_{T(1)}K(C) \rightarrow L_{T(1)}K(A)$. If $f : L_{T(1)}K(A) \rightarrow L_{T(1)}K(C)$ is extension of scalars, the projection formula gives

$$tf = 1 - [L_A]^{-1}.$$

Moreover t is $L_{T(1)}K(A)$ -linear, so it carries $(L_{T(1)}K(C))_{\epsilon_C}$ to $(L_{T(1)}K(A))_{\epsilon_A}$. Equation (5.7) shows that

$$b([L_A])t : (L_{T(1)}K(C))_{\epsilon_C} \longrightarrow (L_{T(1)}K(A))_{\epsilon_A}$$

is a left inverse to $f : (L_{T(1)}K(A))_{\epsilon_A} \rightarrow (L_{T(1)}K(C))_{\epsilon_C}$.

If q is odd, Corollary 5.9 says that $L_{T(1)}K(A) \rightarrow L_{T(1)}K(C)$ is an equivalence. It preserves the transported idempotents, so $(L_{T(1)}K(A))_{\epsilon_A} \rightarrow (L_{T(1)}K(C))_{\epsilon_C}$ is an equivalence as well.

Each map along the finite sequence of Lemma 5.10 is therefore split injective on the complementary averaging summands. Its final value is $L_{T(1)}K(0) = 0$. Their finite composite makes $(L_{T(1)}K(R))_\epsilon$ a retract of zero, whence $(L_{T(1)}K(R))_\epsilon = 0$ and $\epsilon = 0$. Therefore $e = 1$. Since $(1 - [L])e = 0$, this proves $[L] = 1$.

For the more general assertion, replace multiplication by $[L]$ throughout by the automorphism T_L induced by $-\otimes_R L$. Then $T_L^m = \text{id}$; the same idempotents split in the target stable category. Additivity and (5.8) give $tf = \text{id} - T_L^{-1}$, and all other steps are unchanged. This form of the argument does not require a multiplicative structure on the values of F . $\square$

Remark 5.11. The class identity in Theorem 5.2 is in $\pi_0 L_{T(1)}K(R)$. The ordinary rational K_0 identity is a separate condition, as in Theorem 6.1 below.

6. CYCLIC DESCENT OVER RATIONAL RINGS

For a rational finite Galois extension, restriction of scalars defines a transfer on ordinary K -theory by Lemma 2.8. We first recall the criterion that will be applied to the finite free extension of coefficients used to adjoin roots of unity.

Theorem 6.1 (Rational transfer criterion [CMNN20, Theorem 5.6 and Proposition 5.7]). *Let $R \rightarrow S$ be a finite G -Galois extension of rational commutative ring spectra. If*

$$1 \in \mathrm{im}(K_0(S) \otimes \mathbb{Q} \xrightarrow{\mathrm{tr}} K_0(R) \otimes \mathbb{Q}), \quad (6.1)$$

then the canonical map

$$L_{T(1)}K(R) \xrightarrow{\simeq} (L_{T(1)}K(S))^{hG}$$

is an equivalence. Condition (6.1) is equivalent to

$$[S] = |G|[R] \quad \text{in } K_0(R) \otimes \mathbb{Q}. \quad (6.2)$$

Proof. The descent assertion is [CMNN20, Theorem 5.6(2)], applied to the weakly additive invariant $L_{T(1)}K$ in the sense of [CMNN20, Definition 3.10]. This definition does not require preservation of filtered colimits, and $L_1^f L_{T(1)}K = L_{T(1)}K$. Its hypothesis is precisely (6.1); Lemma 2.8 supplies the ordinary perfectness needed for the transfer.

Here is also the elementary equivalence of the two conditions, as in [CMNN20, Proposition 5.7]. Suppose $x \in K_0(S)_{\mathbb{Q}}$ satisfies $\mathrm{tr}(x) = 1$. By the projection formula, $y \mapsto \mathrm{tr}(x \mathrm{res}(y))$ is the identity on $K_0(R)_{\mathbb{Q}}$. Thus restriction is injective. After restriction,

$$\mathrm{res}[S] = [S \otimes_R S] = |G|[S] = \mathrm{res}(|G|[R])$$

in $K_0(S)_{\mathbb{Q}}$, using the Galois product decomposition. Injectivity gives (6.2). Conversely, if (6.2) holds, the transfer of $|G|^{-1}[S]$, with $[S]$ now the unit in $K_0(S)_{\mathbb{Q}}$, is $[R]$. This proves (6.1). $\square$

Remark 6.2. For a rational finite G -Galois extension $R \rightarrow S$, two cases of descent follow directly from existing results. If G is a p -group, [CMNN, Theorem B] applies because $R^{tC_p} = 0$ and $\mathrm{Perf}(R) \simeq \mathrm{Perf}(S)^{hG}$ by Lemma 2.8. If the extension is finite *spectral étale*, meaning that $\pi_0 S$ is finite étale over $\pi_0 R$ and

$$\pi_* R \otimes_{\pi_0 R} \pi_0 S \simeq \pi_* S,$$

then [CMNN20, Proposition 5.4] supplies (6.1), so Theorem 6.1 applies. This includes the case where R is connective: faithfulness follows from Lemma 2.8, and [Mat16, Theorem 6.17] then makes the extension spectral étale.

Proposition 6.3. *Suppose that m is prime to p , that R is a commutative $\mathbb{Q}(\zeta_m)$ -algebra, and that $R \rightarrow S$ is C_m -Galois. Then $L_{T(1)}K(R) \rightarrow (L_{T(1)}K(S))^{hC_m}$ is an equivalence.*

Proof. For $m = 1$ the comparison is the identity, so assume $m > 1$. The character idempotents split S into its eigensummands, with subscripts taken modulo m :

$$S = \bigoplus_{j=0}^{m-1} L_j, \quad L_0 = R.$$

The multiplication maps $L_i \otimes_R L_j \rightarrow L_{i+j}$ are equivalences. To check this, tensor with the faithful algebra S . The Galois isomorphism identifies $S \otimes_R S$ with the functions $C_m \rightarrow S$, on which the group acts by translation. Each character summand is free of rank one, generated by the corresponding invertible character function; their multiplication maps are equivalences. Faithful base change detects the claim.

Thus L_1 is invertible, $L_j \simeq L_1^{\otimes j}$, and $L_1^{\otimes m} \simeq R$. The line is even: its square maps isomorphically to L_2 , and commutativity of S says that this map is unchanged by the self-braiding. Consequently Theorem 5.2 gives

$$[S] = \sum_{j=0}^{m-1} [L_1]^j = m \quad \text{in } \pi_0 L_{T(1)}K(R).$$

Let $f : L_{T(1)}K(R) \rightarrow L_{T(1)}K(S)$ and $t : L_{T(1)}K(S) \rightarrow L_{T(1)}K(R)$ be extension and restriction of scalars. The projection formula gives $tf = m$, while the Galois tensor decomposition gives

$$ft = \sum_{g \in C_m} g.$$

These are the standard coherent induction–restriction formulas; f takes values in invariants and t is invariant. Since m is invertible, homotopy fixed points identify with the averaging summand $m^{-1} \sum g$ of $L_{T(1)}K(S)$. The two formulas show that f identifies $L_{T(1)}K(R)$ with that summand, with inverse $m^{-1}t$. This is precisely the stated comparison map. $\square$

Proposition 6.4. *For every rational C_m -Galois extension with $(m, p) = 1$, the canonical map $L_{T(1)}K(R) \rightarrow (L_{T(1)}K(S))^{hC_m}$ is an equivalence.*

Proof. Let $\Gamma = \text{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$, and form $R' = R \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_m)$ and $S' = S \otimes_{\mathbb{Q}} \mathbb{Q}(\zeta_m)$. The maps $R \rightarrow R'$ and $S \rightarrow S'$ are Γ -Galois. If $d = [\mathbb{Q}(\zeta_m) : \mathbb{Q}]$, then the choice of a $\mathbb{Q}$ -basis of $\mathbb{Q}(\zeta_m)$ gives $R' \simeq R^{\oplus d}$ as R -modules, so $[R'] = d[R]$ already in $K_0(R)$. The transfer of $d^{-1}[R']$, now viewed as a class in $K_0(R') \otimes \mathbb{Q}$, is the unit in $K_0(R) \otimes \mathbb{Q}$. The same holds for $S \rightarrow S'$.

Theorem 6.1 gives the canonical equivalences

$$L_{T(1)}K(R) \simeq (L_{T(1)}K(R'))^{h\Gamma}, \quad L_{T(1)}K(S) \simeq (L_{T(1)}K(S'))^{h\Gamma}.$$

In this application, d may be divisible by p . Proposition 6.3 applies to $R' \rightarrow S'$. Its canonical comparison is Γ -equivariant. Since the two group actions commute, we conclude

$$L_{T(1)}K(R) \simeq ((L_{T(1)}K(S'))^{hC_m})^{h\Gamma} \simeq ((L_{T(1)}K(S'))^{h\Gamma})^{hC_m} \simeq (L_{T(1)}K(S))^{hC_m}.$$

Naturality identifies this composite with the desired comparison. $\square$

Example 6.5 (Periodic root extensions). Let k be a characteristic-zero field containing a primitive m th root of unity ζ . Consider the rational commutative ring spectra associated to the graded commutative dg k -algebras with zero differential

$$\pi_* R = k[v^{\pm 1}], \quad |v| = 2m, \quad \pi_* S = k[u^{\pm 1}], \quad |u| = 2.$$

The map $v \mapsto u^m$, with C_m acting by $u \mapsto \zeta u$, is C_m -Galois. Indeed, rational averaging gives $S^{hC_m} \simeq R$, and

$$S \simeq \bigoplus_{j=0}^{m-1} \Sigma^{2j} R \tag{6.3}$$

with basis $1, u, \dots, u^{m-1}$. Thus S is faithful over R , and the Galois comparison on homotopy groups is

$$k[u^{\pm 1}, w^{\pm 1}]/(w^m - u^m) \simeq \prod_{j=0}^{m-1} k[u^{\pm 1}], \quad w \mapsto (\zeta^j u)_j.$$

The factors are pairwise comaximal since u is a unit. Equation (6.3) gives $[S] = m[R]$ in $K_0(R)$, so Theorem 6.1 yields

$$L_{T(1)}K(R) \xrightarrow{\simeq} (L_{T(1)}K(S))^{hC_m}$$

for every prime p , including those dividing m . For $m > 1$, this extension is not spectral étale: the map $\pi_0 R \rightarrow \pi_0 S$ is the identity of k , but $\pi_2 R = 0$ and $\pi_2 S = k$, so homotopy base change fails. It therefore illustrates a non-spectral-étale Galois extension already covered by the transfer criterion.

7. DESCENT FOR ARBITRARY FINITE GROUPS

Let $\mathcal{F}$ denote the family of cyclic subgroups of G whose orders are prime to p . A genuine G -spectrum M is $\mathcal{F}$ -complete if the canonical map

$$M \longrightarrow F(E\mathcal{F}_+, M)$$

is an equivalence. We use genuine G -spectra whose categorical fixed-point spectra are $T(n+1)$ -local; equivalently, these are spectral Mackey functors valued in $\mathrm{Sp}_{T(n+1)}$. Localization in this category is objectwise on the values G/H .

Proposition 7.1 (Family completeness input). *Let $n \geq 0$, and let A be a commutative ring spectrum such that $L_{T(n)}(A^{tC_p}) = 0$. Let $\mathcal{C}$ be an A -linear small idempotent-complete stable category with an A -linear G -action. Its equivariant K -theory gives a genuine G -spectrum M , localized objectwise, with*

$$M^H = L_{T(n+1)}K(\mathcal{C}^{hH}) \quad (H \leq G). \quad (7.1)$$

Then M is $\mathcal{F}$ -complete.

Proof. The equivariant construction and its multiplicativity are [CMNN, Construction 2.12]. The proof of [CMNN, Theorem 6.5, pp. 27–28] explicitly proves that $L_{T(n+1)}K_G(A)$ is complete for the prime-to- p cyclic family, and that its modules, including (7.1), are complete by [CMNN, Corollary 3.9]. This completeness statement holds for every $n \geq 0$. $\square$

Lemma 7.2 (Detection on a family). *A map $M \rightarrow N$ between $\mathcal{F}$ -complete genuine G -spectra is an equivalence if $M^H \rightarrow N^H$ is an equivalence for every $H \in \mathcal{F}$. Also, a Borel-complete spectrum $F(EG_+, M)$ is $\mathcal{F}$ -complete.*

Proof. Let Q be the cofiber of $M \rightarrow N$. Completeness is preserved by cofibers, since $F(E\mathcal{F}_+, -)$ is exact. The G -cells of $E\mathcal{F}$ have isotropy in $\mathcal{F}$. For every $H \in \mathcal{F}$, restriction of Q to H is zero: all its subgroup fixed points vanish, since $\mathcal{F}$ is closed under subgroups, and categorical fixed points detect equivalences of genuine H -spectra. It follows that $F((G/H)_+, Q) = 0$. Build $E\mathcal{F}$ from its G -cells and use that $F(-, Q)$ takes colimits to limits. This proves $F(E\mathcal{F}_+, Q) = 0$, so completeness gives $Q = 0$.

Finally, $EG \times E\mathcal{F} \simeq EG$ as G -spaces: on the trivial subgroup both are contractible, and on every nontrivial subgroup both have empty fixed-point space. The exponential law gives

$$F(E\mathcal{F}_+, F(EG_+, M)) \simeq F(EG_+, M),$$

proving the last assertion. $\square$

Proof of Theorem 1.2. Apply Proposition 7.1 to $\mathrm{Perf}(S)$ with its R -linear G -action. Since R is $T(n)$ -local, Proposition 2.3(iv) gives $L_{T(n)}(R^{tC_p}) = 0$. Let M be the resulting localized genuine G -spectrum. Both M and its Borel completion $F(EG_+, M)$ are $\mathcal{F}$ -complete, and

$$F(EG_+, M)^H \simeq (L_{T(n+1)}K(S))^{hH}. \quad (7.2)$$

This is the usual fixed-point description of Borel completion; its restriction to H has a free contractible H -space as the indexing space.

For $H \in \mathcal{F}$, Lemma 2.5 makes $S^{hH} \rightarrow S$ an H -Galois extension. Proposition 2.6, together with Proposition 4.1 for $n \geq 1$ and Proposition 6.4 for $n = 0$, gives the natural maps

$$L_{T(n+1)}K(S^{hH}) \xrightarrow{\simeq} L_{T(n+1)}K(\mathrm{Perf}(S)^{hH}) = M^H \longrightarrow F(EG_+, M)^H = (L_{T(n+1)}K(S))^{hH}.$$

The first arrow is an equivalence. The composite is the cyclic descent comparison, hence an equivalence; so the second arrow is also an equivalence. By Lemma 7.2, the canonical map $M \rightarrow F(EG_+, M)$ is an equivalence. Its G -fixed points, together with Proposition 2.6, give

$$L_{T(n+1)}K(R) \xrightarrow{\simeq} L_{T(n+1)}K(\mathrm{Perf}(S)^{hG}) = M^G \xrightarrow{\simeq} F(EG_+, M)^G = (L_{T(n+1)}K(S))^{hG}.$$

The first map is induced by extension of scalars and the second by forgetting equivariant objects with their coherent action. Their composite is the map in (1.2). This completes the proof for every finite G . $\square$

Remark 7.3 (Localized fixed points). For a tame group, averaging realizes invariants and coinvariants as retracts preserved by exact functors. For general G , Proposition 7.1 gives the comparison within genuine spectra with localized fixed-point values; its family-completion limit is formed in that category.

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